

Assignment – Module 1

Overview of Machine Learning

Q1. Introduction to Machine Learning

- a) Define Machine Learning and explain its importance in modern computing.
- b) Differentiate Machine Learning from traditional programming approaches.
- c) Describe the role of data in ML systems and challenges faced during data preparation.

Q2. Types of Machine Learning

- a) Explain Supervised Learning with at least two real-life examples and a workflow diagram.
- b) Explain Unsupervised Learning, including clustering and dimensionality reduction examples.
- c) Explain Reinforcement Learning with a real-world application (e.g., gaming, robotics).
- d) Create a comparative table highlighting the key differences between Supervised, Unsupervised, and Reinforcement Learning.

Q3. Concepts of Regression

- a) Define regression and explain its purpose in ML.
- b) Provide examples of applications where regression is used.
- c) Briefly explain the difference between simple regression and multiple regression.

Q4. Concepts of Classification

- a) Define classification and give examples of real-world classification problems.
- b) Explain the concept of decision boundaries with a neat diagram.
- c) Differentiate between binary and multi-class classification problems.

Q5. Concepts of Clustering

- a) Define clustering and its significance in unsupervised learning.
- b) Explain at least two clustering algorithms (e.g., K-Means, Hierarchical Clustering) with diagrams.
- c) Provide at least two application areas where clustering is widely used.

Q6. Case Study / Application-Based Question

Choose any two real-world problems (e.g., stock price prediction, spam email detection, image recognition, customer segmentation). For each problem:

- Identify which type of ML it represents (Supervised / Unsupervised / RL).
- Specify what would be the input features and expected output.
- Mention which algorithm(s) could be applied and justify your choice.
- Discuss one key challenge in implementing ML for that problem.

Q7. Short Notes

Write short notes (around 100–150 words each) on any three of the following:

- Role of Big Data in Machine Learning
- Importance of Feature Engineering
- Limitations of Machine Learning
- Ethical Concerns and Bias in ML Systems
- Current Trends in Machine Learning (e.g., AutoML, Edge AI)